TRANSFERIQ: DYNAMIC PLAYER TRANSFER VALUE PREDICTION USING AI

AN INFOSYS SPRINGBOARD VIRTUAL INTERNSHIP 6.0 PROJECT

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PROBLEM STATEMENT

- High Spending in Football Transfers
- Subjectivity in Transfer Valuations
- Lack of Transparent Data-Driven Models
- Need for Unified Model: Performance, Market, Sentiment, Injuries

PROJECT OBJECTIVE

Accurate AI-Based Transfer Value Prediction	Predict transfer values, enhancing decision–making in player acquisitions
Multi-Source Data Integration	This project aims to integrate diverse data sources
Instant Forecast for clubs	Develop an accurate, real-time prediction system for club performances
Simple UI Dashboard for Insights	Intuitive data visualization and user-friendly interaction.

DATA SOURCES USED

- **Player Statistics and Performance Metrics**
- **Injury History and Records**
- **Sentiment Data from Fans and Social Media**
- **Market Trends and Past Transfer Values**

SYSTEM ARCHITECTURE

1

Data Collection and Preprocessing

Data is collected from multiple sources. Preprocessing enhance model accuracy and reliability.

2

Feature Engineering and Model Training

To enhance prediction accuracy for transfer value estimation.

3

Model Evaluation and UI Integration

Validated using cross-validation and error metrics and Predictions with intuitive visualization for end-users.

DATA PREPROCESSING

Handling Missing and Duplicate Data

Ensure data accuracy and model reliability in transfer value prediction.

Feature Conversion and Normalization

To ensure uniformity and enhance model performance.

Text Data Cleaning

To ensure accurate and consistent input for the AI model in transfer value prediction.

Model Used: XGBoost

Key Features

- Best for structured player data.
- Handles non-linear patterns effectively.
- Prevents model overfitting.
- Ensures fast training times.

Hyperparameters Tuned

Learning Rate:

0.05

Max Depth:

6

Estimators:

300

Model Performance

Evaluation Metrics: RMSE, MAE, R^2 Score

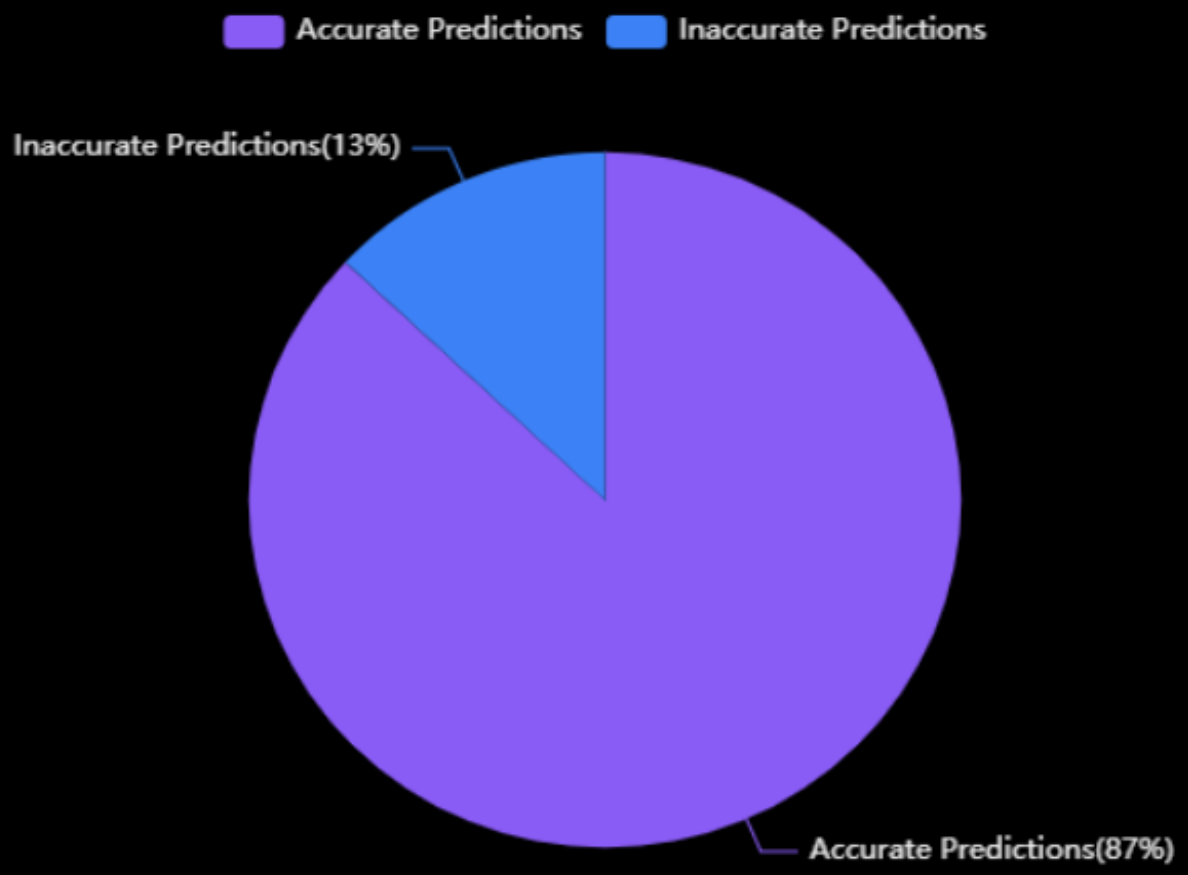
The model's performance is evaluated using RMSE, MAE, and R^2 score

Interpretation of Performance Results

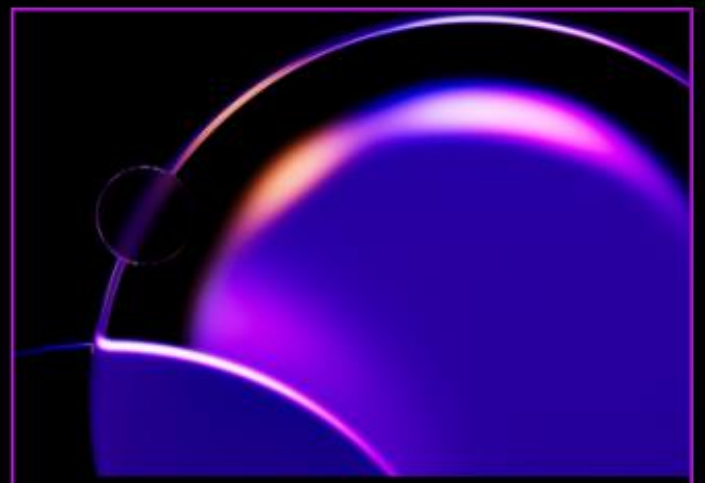
The model demonstrates high accuracy and consistency across diverse datasets

Overall Prediction Accuracy

The model achieves an overall prediction accuracy of 87%,



Final Output and Conclusion

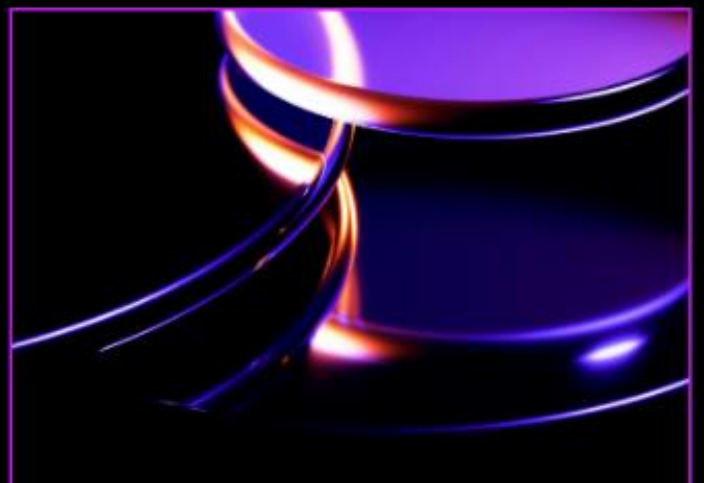


Predicted Transfer Value Display

Enhanced user understanding and support informed decision-making.

Visualization of Trends and Player Insights

Enabled intuitive insights into transfer market dynamics



Summary of Project Impact and Future Scope

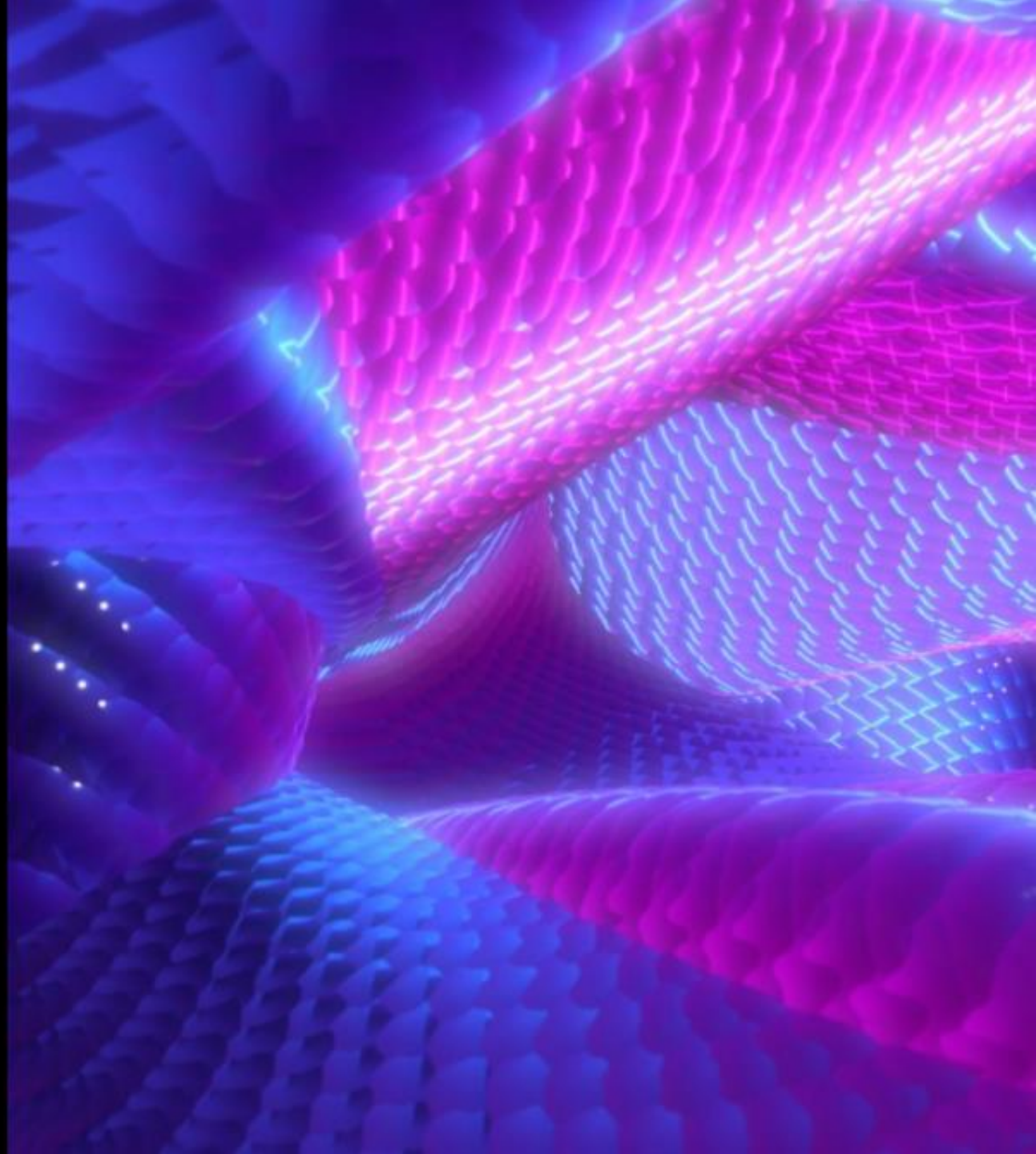
The system enhances transfer decision accuracy, with future potential.

Conclusion

Multi-Source AI Integration

Features Professional UI and Reliable Predictions

Potential Future Enhancements (Deep Learning, Real-Time News)



THANK YOU